

C739

Space, Time and Motion

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher opening a unit on the history of physics, which runs three weeks, describes Galileo's thought experiment: a ball rolls down one ramp and up another. As the second ramp is made less steep, the ball rolls farther. The teacher mentions Galileo also built telescopes. If the second ramp were perfectly flat and frictionless, what did Galileo conclude?

- A. The ball would speed up without limit.
- B. The ball would keep rolling forever at constant speed.
- C. The ball would slow and stop, since rest is its natural state.
- D. It could not be known without a telescope to observe it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A satellite built by a team of 200 engineers is moved from an orbit 1 Earth radius from Earth's center to 3 Earth radii from the center. Its mass is 900 kg. How does Earth's gravitational pull on it change?

- A. It stays the same because the satellite's mass did not change.
- B. It becomes 1/9 as strong.
- C. It becomes 1/3 as strong.
- D. It becomes 9 times as strong.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

The gravitational force between two objects drops to 1/4 of its original value. What happened to the distance between their centers?

- A. It increased by a factor of 1/4.
- B. It doubled.
- C. It was halved.
- D. It quadrupled.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A spaceship with a crew of six moves away from Earth at $0.5c$ and turns on a forward-pointing laser. The ship has been in flight for four months. What speed does an observer on Earth measure for the laser light?

- A. It depends on how long the ship has been flying.
- B. $1.5c$
- C. c
- D. $0.5c$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An astronaut travels at $0.6c$ ($\gamma = 1.25$) on a mission funded by an international space agency. Her onboard clock records 8 years for the trip. How much time passes for mission control on Earth?

- A. 6.4 years
- B. About 13.3 years
- C. 8 years
- D. 10 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A probe travels at $0.8c$ ($\gamma = 5/3$). Observers on Earth record 20 years for its trip. How much time passes on the probe's clock?

- A. 12 years
- B. 20 years
- C. About 33.3 years
- D. 16 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A science teacher who runs the astronomy club says the Sun turns mass into energy. How much energy is released if exactly 1 gram of mass is converted? ($c = 3 \times 10^8$ m/s) The club meets on Thursdays.

- A. 9×10^{16} J
- B. 3×10^{11} J
- C. 3×10^5 J
- D. 9×10^{13} J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a lab using a zinc plate bought from a science catalog, bright red light produces no ejected electrons, but dim ultraviolet light does. The red lamp is rated at 100 watts. What explains this?

- A. Light behaves only as a wave, so energy builds over time.
- B. The red lamp needs more power to eject electrons.
- C. Each UV photon has enough energy (high frequency) to free an electron; red photons do not.
- D. Brighter light always ejects more electrons.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Ultraviolet light already ejects electrons from a metal. Its intensity is doubled while its frequency stays the same. What changes?

- A. The threshold frequency of the metal drops.
- B. No change at all.
- C. About twice as many electrons are ejected, with the same maximum kinetic energy.
- D. The electrons' maximum kinetic energy doubles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Two identical atomic clocks, each costing about \$5,000, are placed one at sea level and one on a mountain 3,000 m higher. After a year, compared side by side, which has ticked more?

- A. The mountain clock
- B. The sea-level clock
- C. The mountain clock, because it moves faster with Earth's rotation
- D. Neither; identical clocks always agree.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Using $H_0 = 70$ km/s per megaparsec, a class studies a galaxy 100 megaparsecs away that was photographed by a space telescope launched in 1990. What is its predicted recession speed?

- A. 7,000 km/s
- B. 70 km/s
- C. 0.7 km/s
- D. The same as every other galaxy

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Using $H_0 = 70$ km/s per megaparsec, a galaxy is receding at 14,000 km/s. About how far away is it?

- A. 14,000 megaparsecs
- B. It cannot be estimated from speed.
- C. 980,000 megaparsecs
- D. 200 megaparsecs

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A teacher opening a unit on the history of physics, ~~which runs three weeks~~, describes Galileo's thought experiment: a ball rolls down one ramp and up another. As the second ramp is made less steep, the ball rolls farther. ~~The teacher mentions Galileo also built telescopes.~~ If the second ramp were perfectly flat and frictionless, what did Galileo conclude?

Target word: Galileo's conclusion **Decides:** No friction and no slope means no cause to stop.

A	Inverted operation	Speeding up needs a net force.
B	Correct	Galileo's idea of inertia.
C	Confusion set	That is Aristotle's view.
D	Camouflage	The telescope detail does not bear on the thought experiment.

2. Answer B Concept 2

A satellite ~~built by a team of 200 engineers~~ is moved from an orbit 1 Earth radius from Earth's center to 3 Earth radii from the center. ~~Its mass is 900 kg.~~ How does Earth's gravitational pull on it change?

Target word: force ratio **Decides:** Distance tripled: $1/3^2 = 1/9$.

A	Absolute language	Force depends on distance as well as mass.
B	Correct	Inverse square: $1/(3^2)$.
C	Dropped element	Leaves out the square.
D	Inverted operation	Force falls, not rises, with distance.

3. Answer B Concept 2 · transfer

The gravitational force between two objects drops to 1/4 of its original value. What happened to the distance between their centers?

Target word: solve for distance change **Decides:** $1/r^2 = 1/4$ means r doubled.

A	Wrong stem number	Reuses the force ratio as the distance ratio.
B	Correct	$2^2 = 4$.
C	Inverted operation	Halving the distance quadruples the force.
D	Dropped element	Would make the force 1/16.

4. Answer C Concept 3

A spaceship ~~with a crew of six~~ moves away from Earth at 0.5c and turns on a forward-pointing laser. ~~The ship has been in flight for four months.~~ What speed does an observer on Earth measure for the laser light?

Target word: measured speed of light **Decides:** Light speed is the same for all inertial observers.

A	Camouflage	Flight time does not change the measured speed of light.
B	Confusion set	Galilean velocity addition does not apply to light.
C	Correct	Einstein's second postulate.
D	Inverted operation	Subtracts the ship's speed.

5. Answer D Concept 4

An astronaut travels at $0.6c$ ($\gamma = 1.25$) on a mission ~~funded by an international space agency~~. Her onboard clock records 8 years for the trip. How much time passes for mission control on Earth?

Target word: Earth time **Decides:** Earth time = $\gamma \times$ proper time.

A	Inverted operation	Divides by γ ; the traveler's time is the shorter one.
B	Confusion set	Divides by the speed fraction (0.6) instead of using γ .
C	Absolute language	Assumes time passes the same for all observers.
D	Correct	$1.25 \times 8 = 10$ years.

6. Answer A Concept 4 • transfer

A probe travels at $0.8c$ ($\gamma = 5/3$). Observers on Earth record 20 years for its trip. How much time passes on the probe's clock?

Target word: solve for proper time **Decides:** Proper time = Earth time / γ .

A	Correct	$20 / (5/3) = 12$ years.
B	Wrong stem number	The Earth time, unchanged.
C	Inverted operation	Multiplies by γ .
D	Confusion set	Multiplies by the speed fraction 0.8.

7. Answer D Concept 5

A science teacher ~~who runs the astronomy club~~ says the Sun turns mass into energy. How much energy is released if exactly 1 gram of mass is converted? ($c = 3 \times 10^8$ m/s) ~~The club meets on Thursdays.~~

Target word: energy from 1 g **Decides:** $E = (0.001 \text{ kg})(9 \times 10^{16} \text{ m}^2/\text{s}^2)$.

A	Dropped element	Leaves the mass as 1 instead of 0.001 kg.
B	Inverted operation	Multiplies c by 1,000 instead of dividing the mass.
C	Confusion set	Uses c instead of c^2 .
D	Correct	$0.001 \times 9 \times 10^{16} = 9 \times 10^{13}$ J.

8. Answer C Concept 6

In a lab using a zinc plate ~~bought from a science catalog~~, bright red light produces no ejected electrons, but dim ultraviolet light does. ~~The red lamp is rated at 100 watts.~~ What explains this?

Target word: why UV works and red does not **Decides:** Photon energy depends on frequency.

A	Confusion set	The classical wave view the photoelectric effect disproved.
B	Camouflage	More power adds photons, not energy per photon.
C	Correct	$E = hf$ sets whether any single photon can eject an electron.
D	Absolute language	Only above the threshold frequency.

9. Answer C Concept 6 · transfer

Ultraviolet light already ejects electrons from a metal. Its intensity is doubled while its frequency stays the same. What changes?

Target word: effect of intensity above threshold **Decides:** More photons, same energy per photon.

A	Adjacent category	The threshold is a property of the metal.
B	Absolute language	More photons free more electrons.
C	Correct	Intensity sets photon count; frequency sets energy per photon.
D	Inverted operation	That would require higher frequency.

10. Answer A Concept 7

Two identical atomic clocks, ~~each costing about \$5,000~~, are placed one at sea level and one on a mountain 3,000 m higher. After a year, compared side by side, which has ticked more?

Target word: which clock runs faster **Decides:** Clocks deeper in gravity run slower.

A	Correct	It sits in slightly weaker gravity, so it runs slightly faster.
B	Inverted operation	Stronger gravity slows clocks.
C	Adjacent category	Special-relativity motion effects would slow it; the gravitational effect is the one that decides here.
D	Absolute language	General relativity predicts, and experiments confirm, a difference.

11. Answer A Concept 8

Using $H_0 = 70$ km/s per megaparsec, a class studies a galaxy 100 megaparsecs away ~~that was photographed by a space telescope launched in 1990~~. What is its predicted recession speed?

Target word: recession speed **Decides:** $v = H_0 \times d$.

A	Correct	$70 \times 100 = 7,000$ km/s.
B	Wrong stem number	The constant, not the speed.
C	Inverted operation	Divides H_0 by distance.
D	Absolute language	Speed grows with distance.

12. Answer D Concept 8 · transfer

Using $H_0 = 70$ km/s per megaparsec, a galaxy is receding at 14,000 km/s. About how far away is it?

Target word: solve for distance **Decides:** $d = v / H_0$.

A	Wrong stem number	Reuses the speed as a distance.
B	Dropped element	Hubble's law links speed to distance directly.
C	Inverted operation	Multiplies instead of dividing.
D	Correct	$14,000 / 70 = 200$ Mpc.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Galileo's inertia vs. Aristotle's natural motion	Galileo argued that a moving object on a frictionless level surface keeps moving forever; Aristotle held that motion needs a continuing cause.	Reading "things naturally slow down" as a law rather than an effect of friction.	A thought experiment with a frictionless plane or ramp.
2	Newton's inverse-square gravitation	Gravitational force varies as $1/r^2$: doubling the distance cuts the force to one-fourth.	Scaling force as $1/r$, or measuring distance from the surface instead of the center.	Distance multiplied by a factor and a question about force.
3	Constancy of the speed of light	Every inertial observer measures light in a vacuum at c , regardless of the motion of the source or observer.	Adding velocities the Galilean way to light.	A moving source emitting light.
4	Time dilation	A clock moving relative to an observer runs slow by $\gamma = 1/\sqrt{1 - v^2/c^2}$; the traveler's own (proper) time is the shortest.	Dividing by γ in the wrong place, or assuming time is the same for all observers.	A speed given as a fraction of c and two different clocks.
5	Mass-energy equivalence	$E = mc^2$, with m in kilograms and $c = 3 \times 10^8$ m/s; a small mass holds enormous energy, which powers stars.	Leaving mass in grams or using c instead of c^2 .	A mass converted to energy, fusion, or the Sun.
6	Photon hypothesis (photoelectric effect)	Light comes in photons of energy $E = hf$; electrons are ejected only if the photon frequency exceeds the metal's threshold, regardless of brightness.	Thinking brighter light of any color will eventually eject electrons.	Light intensity vs. color (frequency) on a metal.
7	General relativity: gravity and time	Gravity is the curvature of spacetime; clocks deeper in a gravitational field run slower (gravitational time dilation).	Thinking stronger gravity speeds clocks, or that relativity ignores gravity.	Clocks at different altitudes, GPS satellites, or near a black hole.
8	Hubble's law and cosmic expansion	Recession speed is proportional to distance, $v = H_0 d$ ($H_0 \approx 70$ km/s per megaparsec), evidence the universe is expanding.	Thinking galaxies move away from us at the center, or that speed is the same for all galaxies.	A galaxy distance and a recession speed.

Course scope (public description): This course begins with a quick tour of discovery and exploration in physics, from the ancient Greek philosophers on to Galileo Galilei, Isaac Newton, and Albert Einstein. Einstein's work then serves as the departure point for a detailed look at the properties of motion, time, space, matter, and energy. The course considers Einstein's special theory of relativity, his photon hypothesis, wave-particle duality, his general theory of relativity and its implications for astrophysics and cosmology, as well as his three-decade quest for a unified field theory. It also looks at Einstein as a social and political figure and his contributions as a social and political force. Scientist-authored essays, online interaction, videos, and web resources enable learners to trace this historic path of discovery and explore implications of technology for society, energy production in stars, black holes, the Big Bang, and the role of the scientist in modern society.